

EC2 daily task

Date:25/08/26

1. Launch one EC2 using Amazon Linux 2 image and add a script in user data to install Apache.

Objective :To launch an Amazon EC2 instance and automatically install and configure the Apache HTTP Server using EC2 User Data.

Requirements

AWS Account

Amazon EC2

Amazon Linux 2 AMI

t2.micro instance

Key pair (.pem)

Security Group

Procedure

1. Launch an EC2 Instance

Open the AWS Management Console.

Navigate to EC2 → Instances.

Click Launch instances.

Enter the instance name:

apache-userdata-server

Select Amazon Linux 2 AMI.

Select the t2.micro instance type.

Select or create a key pair.

2. Configure the Security Group

Create a new security group with the following inbound rules:

Type

Protocol

Port

Source

SSH

TCP

22

My IP

HTTP

TCP

80

0.0.0.0/0

SSH is used to connect to the EC2 instance, while HTTP is required to access the Apache web server from a browser.

3. Configure User Data

Go to Advanced details → User data and enter the following script:

```
#!/bin/bash
yum update -y
yum install -y httpd
systemctl enable httpd
systemctl start httpd
echo "<h1>Apache Installed Successfully using User Data</h1>" >
/var/www/html/index.html
```

4. Launch the Instance

Click Launch instance and wait until:

Instance state shows Running

Status checks show 2/2 checks passed

5. Connect to the EC2 Instance

Store the .pem file in a separate directory and connect using Git Bash:

```
ssh -i ~/awk-key/ec2-task-key.pem ec2-user@YOUR_PUBLIC_IP
```

Replace YOUR_PUBLIC_IP with the Public IPv4 address of the EC2 instance.

6. Verify Apache

After connecting to the instance, check the Apache service:

```
sudo systemctl status httpd
```

Expected result:

Active: active (running)

7. Verify the HTML File

Run:

```
cat /var/www/html/index.html
```

Expected output:

Apache Installed Successfully using User Data

8. Test Using a Browser

Open the following URL in a web browser:

http://YOUR_PUBLIC_IP

The following message should be displayed:

EC2 > Instances > Launch an instance

Success
Successfully initiated launch of instance (i-0ffe7d60a702eb4a3)

Launch log

Next Steps

What would you like to do next with this instance, for example "create alarm" or "create backup"

< 1 2 3 4 5 6 >

Create billing usage alerts

To manage costs and avoid surprise bills, set up email notifications for billing usage thresholds.

Create billing alerts

Connect to your instance

Once your instance is running, log into it from your local computer.

Connect to instance

Connect an RDS database

Configure the connection between an EC2 instance and a database to allow traffic flow between them.

Connect an RDS database

Create EBS snapshot policy

Create a policy that automates the creation, retention, and deletion of EBS snapshots

Create EBS snapshot policy

EC2 > Instances > i-0ffe7d60a702eb4a3

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Instance summary for i-0ffe7d60a702eb4a3 (apache-userdata-server) Info

Connect Instance state Actions

Updated less than a minute ago

Instance ID
i-0ffe7d60a702eb4a3

IPv6 address

-

Hostname type
IP name: ip-172-31-91-35.ec2.internal

Answer private resource DNS name
IPv4 (A)

Auto-assigned IP address
44.201.192.167 [Public IP]

Public IPv4 address
44.201.192.167 | open address

Instance state
Running

Private IP DNS name (IPv4 only)
ip-172-31-91-35.ec2.internal

Instance type
t2.micro

VPC ID
vpc-0b174515696de2fd5

Private IPv4 addresses
172.31.91.35

Public DNS
ec2-44-201-192-167.compute-1.amazonaws.com | open address

Elastic IP addresses
-

AWS Compute Optimizer finding
Opt-in to AWS Compute Optimizer for reco

Objective : To launch an EC2 instance using an Ubuntu image and automatically install and configure the Nginx web server using EC2 User Data.

Procedure

1. Launch EC2 Instance

Log in to the AWS Management Console.

Navigate to EC2 → Instances → Launch instances.

Enter the instance name:

ubuntu-nginx-userdata

Under Application and OS Images (AMI), select Ubuntu Server LTS.

Select the t2.micro instance type.

Select the required key pair.

2. Configure Security Group

Create a security group with the following inbound rules:

Type

Protocol

Port

Source

SSH

TCP

22

My IP

HTTP

TCP

80

0.0.0.0/0

SSH (22) is used to connect to the Ubuntu instance.

HTTP (80) is required to access the Nginx web server.

3. Configure User Data

Go to Advanced details → User data and enter:

```
#!/bin/bash
```

```
apt update -y
```

```
apt install -y nginx
```

```
systemctl enable nginx
```

```
systemctl start nginx
```

```
echo "<h1>Nginx Installed Successfully using User Data</h1>" > /var/www/html/index.html
```

4. Launch the Instance

Click Launch instance.

Wait until:

Instance state: Running

Status checks: 2/2 checks passed

5. Connect to Ubuntu Instance

Copy the Public IPv4 address of the instance.

Using Git Bash:

```
ssh -i ~/awk-key/ec2-task-key.pem ubuntu@YOUR_PUBLIC_IP
```

The username for Ubuntu is:

ubuntu

6. Verify Nginx Service

After connecting to the instance, run:

```
sudo systemctl status nginx
```

Expected output:

Active: active (running)

This confirms that Nginx was successfully installed and started.

7. Verify the User Data HTML File

Run:

```
cat /var/www/html/index.html
```

Expected output:

Nginx Installed Successfully using User Data

8. Verify Using Browser

Copy the Ubuntu EC2 instance's Public IPv4 address and open:

http://YOUR_PUBLIC_IP

EC2 > Instances > Launch an instance

Success
Successfully initiated launch of instance (i-0b57adda2b93c4624)

Launch log

Next Steps

What would you like to do next with this instance, for example "create alarm" or "create backup"

< 1 2 3 4 5 6 >

Create billing usage alerts

To manage costs and avoid surprise bills, set up email notifications for billing usage thresholds.

Create billing alerts

Connect to your instance

Once your instance is running, log into it from your local computer.

Connect to instance

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Connect an RDS database

Create EBS snapshot policy

Create a policy that automates the creation, retention, and deletion of EBS snapshots

Create EBS snapshot policy

EC2 > Instances > i-0b57adda2b93c4624

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Instance summary for i-0b57adda2b93c4624 (ubuntu-nginx-userdata) Info

Connect Instance state Actions

Updated less than a minute ago

Instance ID
i-0b57adda2b93c4624

IPv6 address
-

Hostname type
IP name: ip-172-31-92-45.ec2.internal

Answer private resource DNS name
IPv4 (A)

Auto-assigned IP address
3.82.148.224 [Public IP]

Public IPv4 address
3.82.148.224 | open address

Instance state
Running

Private IP DNS name (IPv4 only)
ip-172-31-92-45.ec2.internal

Instance type
t2.micro

VPC ID
vpc-0b174515695de2fd5

Private IPv4 addresses
172.31.92.45

Public DNS
ec2-3-82-148-224.compute-1.amazonaws.com | open address

Elastic IP addresses
-

AWS Compute Optimizer finding
Opt-in to AWS Compute Optimizer for reco

```
ubuntu@ip-172-31-92-45: ~
chivka@karthik: MINGW64 ~/awk-key
$ ssh -i ~/awk-key/ec2-task-key.pem ubuntu@3.82.148.224
The authenticity of host '3.82.148.224 (3.82.148.224)' can't be established.
ED25519 key fingerprint is SHA256:b1b21TCoX5MKnTBPR4cfbpUYxzdxWYH0o+R7kTq2JI.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '3.82.148.224' (ED25519) to the list of known hosts.
Welcome to Ubuntu 20.04 LTS (GNU/Linux 7.0.0-1006-aws x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Tue Aug 25 10:33:06 UTC 2026

System load: 0.16          Processes: 115
Usage of /: 34.0% of 6.61GB Users logged in: 0
Memory usage: 25%          IPv4 address for enx0: 172.31.92.45
Swap usage: 0%

Expanded Security Maintenance for Applications is not enabled.

120 updates can be applied immediately.
90 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

ubuntu@ip-172-31-92-45:~$ sudo systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
   Active: active (running) since Tue 2026-08-25 10:25:15 UTC; 8min ago
   Invocation: 54b8faec38254d018328d80eac6163e4
      Docs: man:nginx(8)
    Main PID: 1494 (nginx)
       Tasks: 2 (limit: 657)
      Memory: 2.3M (peak: 5.1M)
         CPU: 27ms
    CGroup: /system.slice/nginx.service
            └─1494 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
              └─1497 "nginx: worker process"

Aug 25 10:25:15 ip-172-31-92-45 systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy server...
Aug 25 10:25:15 ip-172-31-92-45 systemd[1]: Started nginx.service - A high performance web server and a reverse proxy server.
ubuntu@ip-172-31-92-45:~$ cat /var/www/html/index.html
<h1>Nginx Installed Successfully using User Data</h1>
```

Nginx Installed Successfully using User Data

3. Launch one Windows server and install Tomcat on Windows.

Objective : To launch a Windows Server EC2 instance and install Apache Tomcat on the Windows server. The Tomcat server was configured and verified successfully on port 8080.

Step 1: Launch Windows EC2 Instance

Logged in to the AWS Management Console.

Navigated to EC2 → Instances → Launch Instance.

Entered the instance name:

windows-tomcat-server

Selected Microsoft Windows Server 2022 Base as the AMI.

Selected the required instance type.

Selected the existing key pair:

ec2-task-key

Configured the Security Group with the required inbound rules.

Type

Protocol

Port

Source

RDP

TCP

3389

My IP

Custom TCP

TCP

8080

0.0.0.0/0

RDP port 3389 was used to connect to the Windows server, and port 8080 was used for Tomcat.

Launched the instance.

Step 2: Verify Instance

After launching, verified that:

Instance State: Running

Status Checks: 2/2 passed

The Windows EC2 instance was successfully running.

Step 3: Retrieve Windows Administrator Password

Selected the Windows EC2 instance.

Clicked Connect.

Selected RDP Client.

Clicked Get Password.

Uploaded the private key:

ec2-task-key.pem

Clicked Decrypt Password.

Retrieved the Windows Administrator credentials.

Username:

Administrator

Step 4: Connect to Windows Server

Downloaded the RDP shortcut file.

Opened the .rdp file.

Connected using:

Username: Administrator

Password: Decrypted Windows password

Successfully logged in to the Windows EC2 server.

Step 5: Verify Java

Opened Command Prompt and executed:

```
java -version
```

Initially, Java was not installed. Therefore, JDK was installed.

After installation, verified Java using:

```
java -version
```

and:

```
javac -version
```

Java was successfully installed and available from the command line.

Step 6: Download Apache Tomcat

Opened the Apache Tomcat official website.

Selected Tomcat 10.1.

Downloaded the 64-bit Windows ZIP / Windows Service Installer.

Used Apache Tomcat 10.1.59.

Step 7: Install Apache Tomcat

The Tomcat Windows Service Installer was used.

The installation directory was configured as:

C:\Tomcat

Tomcat configuration:

HTTP/1.1 Connector Port: 8080

The installation was completed successfully.

Step 8: Verify Tomcat Service

Opened Command Prompt and executed:

```
sc query TOMCAT10
```

The output showed:

```
SERVICE_NAME: TOMCAT10
```

```
STATE: 4 RUNNING
```

This confirmed that the Tomcat service was running successfully.

Step 9: Verify Tomcat Port

Executed:

```
netstat -ano | findstr :8080
```

The output showed:

```
TCP 0.0.0.0:8080 LISTENING
```

```
TCP [::]:8080 LISTENING
```

This confirmed that Tomcat was listening on port 8080.

Step 10: Configure Windows Firewall

To allow external connections to Tomcat port 8080, executed the following command from an Administrator Command Prompt:

```
netsh advfirewall firewall add rule name="Tomcat 8080" dir=in action=allow protocol=TCP localport=8080
```

The command completed successfully.

Step 11: Test Tomcat Locally

Opened a browser on the Windows EC2 server and accessed:

```
http://localhost:8080
```

The Apache Tomcat welcome page was displayed:

```
Apache Tomcat/10.1.59
```

If you're seeing this, you've successfully installed Tomcat.

Congratulations!

Step 12: External Browser Verification

The Windows EC2 instance had the following public IP:

```
54.164.200.60
```

From the local computer, Tomcat was accessed using:

```
http://54.164.200.60:8080
```

The Tomcat welcome page was successfully displayed.

us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#LaunchInstances:

aws

Search

[Alt+S] Ask Amazon Q

United States (N. Virginia)

CHVK (634176595009)

EC2 > Instances > Launch an instance

Success

Successfully initiated launch of instance (i-09bdcac2335857d8b)

Launch log

Next Steps

What would you like to do next with this instance, for example "create alarm" or "create backup"

1 2 3 4 5 6 >

Create billing usage alerts

To manage costs and avoid surprise bills, set up email notifications for billing usage thresholds.

Create billing alerts

Connect to your instance

Once your instance is running, log into it from your local computer.

Connect to instance

Connect an RDS database

Configure the connection between an EC2 instance and a database to allow traffic flow between them.

Connect an RDS database

Create EBS snapshot policy

Create a policy that automates the creation, retention, and deletion of EBS snapshots

Create EBS snapshot policy

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us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#InstanceDetails:instanceId=i-09bdcac2335857d8b

aws

Search

[Alt+S] Ask Amazon Q

United States (N. Virginia)

CHVK (634176595009)

EC2 > Instances > i-09bdcac2335857d8b

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Instance summary for i-09bdcac2335857d8b (windows-tomcat-server)

Info

Connect Instance state Actions

Updated less than a minute ago

Instance ID

i-09bdcac2335857d8b

Public IPv4 address

54.164.200.60 | open address

Private IPv4 addresses

172.31.92.9

IPv6 address

-

Instance state

Running

Public DNS

ec2-54-164-200-60.compute-1.amazonaws.com | open address

Hostname type

IP name: ip-172-31-92-9.ec2.internal

Private IP DNS name (IPv4 only)

ip-172-31-92-9.ec2.internal

Answer private resource DNS name

IPv4 (A)

Instance type

t2.micro

Auto-assigned IP address

54.164.200.60 (Public IP)

VPC ID

vpc-0b174515696de2fd5

Elastic IP addresses

-

AWS Compute Optimizer finding

Opt-in to AWS Compute Optimizer for recommendations

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Find Help

Apache Tomcat/10.1.59

If you're seeing this, you've successfully installed Tomcat. Congratulations!

Recommended Reading:

[Security Considerations How-To](#)
[Manager Application How-To](#)
[Clustering/Session Replication How-To](#)

Server Status

Manager App

Host Manager

Developer Quick Start

[Tomcat Setup](#)
[First Web Application](#)

[Realms & AAA](#)
[JDBC DataSources](#)

[Examples](#)

[Servlet Specifications](#)
[Tomcat Versions](#)

Managing Tomcat

For security, access to the [manager webapp](#) is restricted. Users are defined in: `$CATALINA_HOME/conf/tomcat-users.xml`

In Tomcat 10.1 access to the manager application is split between different users. [Read more...](#)

Documentation

[Tomcat 10.1 Documentation](#)
[Tomcat 10.1 Configuration](#)

[Tomcat Wiki](#)

Find additional important configuration information in:

Getting Help

[FAQ and Mailing Lists](#)

The following mailing lists are available:

[tomcat-announce](#)

Important announcements, releases, security vulnerability notifications. (Low volume).

[tomcat-users](#)

4. Take a snapshot of the instance created in Task 1.

Objective : To create an EBS snapshot of the Amazon Linux 2 EC2 instance created in Task

Step 1: Select the Task 1 EC2 Instance

Logged in to the AWS Management Console.

Navigated to EC2 → Instances.

Selected the instance created in Task 1:

Instance Name: apache-userdata-server

Instance ID: i-0ffe7d60a702eb4a3

Verified that the instance was in the Running state.

Step 2: Identify the Root EBS Volume

Opened the Storage section of the EC2 instance.

Identified the root EBS volume.

The volume details were:

Volume ID: vol-0e5a31a26c55a2cbf

Volume Type: gp2

Size: 8 GiB

Step 3: Create the Snapshot

Navigated to:

EC2 → Elastic Block Store → Volumes

Selected:

vol-0e5a31a26c55a2cbf

Clicked Actions → Create snapshot.

Entered the description:

Task 1 Apache EC2 Snapshot

Clicked Create snapshot.

Step 4: Verify the Snapshot

Navigated to:

EC2 → Elastic Block Store → Snapshots

Located the newly created snapshot.

Initially, the snapshot status was:

Pending

Waited until the snapshot status changed to:

Completed

us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#Volumes:volumeId=vol-0e5a31a26c55a2cbf

aws [Search] [Alt+S] United States (N. Virginia) CHVK (634176595009) CHVK

Capacity reservations
Capacity Manager New

▼ Images
AMIs
AMI Catalog

▼ Elastic Block Store
Volumes
Snapshots
Lifecycle Manager

▼ Network & Security
Security Groups
Elastic IPs
Placement Groups
Key Pairs
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▼ Load Balancing

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Successfully created snapshot snap-050b40bfa3aaf0387 from volume vol-0e5a31a26c55a2cbf.
If you need your snapshot to be immediately available consider using Fast Snapshot Restore. Manage fast snapshot restore

Volumes (1) Info Last updated less than a minute ago Recycle Bin Actions Create volume

Saved filter sets
Choose filter set Search
Volume ID = vol-0e5a31a26c55a2cbf Clear filters

Name Volume ID Type Size IOPS Throughput Snapshot ID

Fault tolerance for all volumes in this Region

Snapshot summary Last updated on Tue, Aug 25, 2026, 05:08:00 PM (GMT+05:30)

Snapshots to back up volumes can be manually created, or created through automation such as Amazon Data Lifecycle Manager, AWS Backup, custom scripts, or third-party tools.

Volumes backed up in the last 24 hours / Total # volumes Data Lifecycle Manager default policy for EBS Snapshots status

us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#Snapshots:

aws [Search] [Alt+S] Ask Amazon Q United States (N. Virginia) CHVK (634176595009) CHVK

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Snapshots (1) Info Last updated less than a minute ago Recycle Bin Actions Create snapshot

Snapshot scope Owned by me Search

Snapshot ID Full snapshot size Volume size Description Storage tier Snapshot status

nap-050b40bfa3aaf0387 - 8 GiB Task 1 Apache EC2 Snapshot Standard Pending

Select a snapshot above.

us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#Snapshots:

aws [Search] [Alt+S] Ask Amazon Q United States (N. Virginia) CHVK (634176595009) CHVK

Capacity reservations
Capacity Manager New

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AMIs
AMI Catalog

▼ Elastic Block Store
Volumes
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Key Pairs
Network Interfaces

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Snapshots (1) Info Last updated less than a minute ago Recycle Bin Actions Create snapshot

Snapshot scope Owned by me Search

Snapshot ID Full snapshot size Volume size Description Storage tier Snapshot status

nap-050b40bfa3aaf0387 3.72 GiB 8 GiB Task 1 Apache EC2 Snapshot Standard Completed

Select a snapshot above.

5. Assign passwordless authentication for the EC2 created in Task 2.

Objective: To configure passwordless SSH authentication for the Ubuntu EC2 instance created in Task 2

Step 1: Connect to the Ubuntu EC2 Instance

The Ubuntu EC2 instance was accessed using the AWS key pair:

ec2-task-key.pem

Command used:

```
ssh -i ~/aws-key/ec2-task-key.pem ubuntu@3.82.148.224
```

The connection was successful.

Step 2: Generate SSH Key Pair on Local PC

On the local PC, Git Bash was opened and the following command was executed:

```
ssh-keygen -t rsa
```

The default location was selected by pressing Enter.

The generated keys were:

~/.ssh/id_rsa

~/.ssh/id_rsa.pub

id_rsa → Private key

id_rsa.pub → Public key

Step 3: Verify the Public Key

The public key was displayed using:

```
cat ~/.ssh/id_rsa.pub
```

The generated public key was copied.

Step 4: Configure SSH Directory on Ubuntu

After connecting to the Ubuntu EC2 instance, the .ssh directory was created:

```
mkdir -p ~/.ssh
```

```
chmod 700 ~/.ssh
```

Step 5: Add Public Key to Ubuntu

The SSH authorized keys file was opened:

```
nano ~/.ssh/authorized_keys
```

The public key generated on the local PC was pasted into this file.

The file was saved using:

Ctrl + O

Enter

Ctrl + X

Step 6: Set Required Permissions

The following permissions were configured:

```
chmod 600 ~/.ssh/authorized_keys
```

Ownership was also verified/configured:

```
chown -R ubuntu:ubuntu ~/.ssh
```

Step 7: Test Passwordless Authentication

The SSH session was exited:

exit

From the local PC, the Ubuntu server was accessed using the newly generated private key:

```
ssh -i ~/.ssh/id_rsa ubuntu@3.82.148.224
```

The connection was established successfully without using the original EC2 .pem key.

```
chvka@karthik MINGW64 ~/awk-key
$ ssh -i ~/.ssh-key/ec2-task-key.pem ubuntu@3.82.148.224
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-1006-aws x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

System information as of Tue Aug 25 12:15:06 UTC 2026

System load:  0.12      Processes:      118
Usage of /:   34.4% of 6.61GB    Users logged in: 0
Memory usage: 26%          IPv4 address for enx0: 172.31.92.45
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

120 updates can be applied immediately.
90 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Tue Aug 25 12:09:37 2026 from 103.160.27.199
ubuntu@ip-172-31-92-45:~$ mkdir -p ~/.ssh
ubuntu@ip-172-31-92-45:~$ chmod 700 ~/.ssh
ubuntu@ip-172-31-92-45:~$ nano ~/.ssh/authorized_keys

chvka@karthik MINGW64 ~/awk-key
$ cat ~/.ssh/id_rsa.pub
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQGCORrNLFM3qGg9prKujbA8xDnrdCUX8GRxnHUGRb5F6Tcz0+hFoDqZnORbGNGoFbctTTAGbEdfGB4qnm5mkcLZ66AqEj29jc/tjn1quU89ng9CBibGqf2IFUwhwSF/k4XTDr
pOQB0F1KD3+HB82RKJjnnto7rwcGgP59GsZk5FKntGp04GMNj09cdm0x1s1dhY7a9eCg5evpgwa4ViTuMqRTuvvtzJ2EaYavFmFK1aLMP/Sy8At2YRFcthoUrd0MzHucTE/+ho31XTBcdWwy4BzwwcEmkzr6bKZBO20z/1MOZNG8V
54uPRt/D6okE1vx1hhd1ytOPaR6YQ0N8xmg/ej1qCrtROMHQAQCJoqIGot2pwgDh2UDG1anueR5Ug+d08JmqjFncG5Bm3IA2QnqXBO7rxK4PD3TnnRpQFTE1oOGE8FuJ10A+3X1FTRBt7EjksRVQLw05ZM4VkuXRRSP4TVYeRC
/qwDonsyA85qw6Mg8YnjMNOvnbA9fajh16= chvka@karthik

ubuntu@ip-172-31-92-45:~$ chmod 600 ~/.ssh/authorized_keys
ubuntu@ip-172-31-92-45:~$ chown -R ubuntu:ubuntu ~/.ssh
ubuntu@ip-172-31-92-45:~$ exit
logout
Connection to 3.82.148.224 closed.

chvka@karthik MINGW64 ~/awk-key
$ ssh -i ~/.ssh/id_rsa ubuntu@3.82.148.224
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-1006-aws x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

System information as of Tue Aug 25 12:18:54 UTC 2026

System load:  0.1      Processes:      117
Usage of /:   34.4% of 6.61GB    Users logged in: 0
Memory usage: 26%          IPv4 address for enx0: 172.31.92.45
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

120 updates can be applied immediately.
90 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Tue Aug 25 12:15:06 2026 from 103.160.27.199
ubuntu@ip-172-31-92-45:~$ |
```

6. Launch any EC2 using the spot purchasing option.

Objective :To launch an EC2 instance using the AWS Spot purchasing option.

Steps Performed

Navigated to AWS Console → EC2 → Launch Instance.

Selected an Amazon Linux 2 x86_64 AMI.

Selected instance type:

t2.micro

Selected the required key pair.

Enabled the Spot purchasing option.

Provided the required network and security group configuration.

Launched the instance with the name:

spot-ec2-server

Verified the instance under EC2 → Instances.

The instance was successfully launched and reached:

Instance State: Running

Navigated to EC2 → Spot Requests and verified the Spot request.

Verification

The Spot Request showed:

Request Type: Instance

Instance Type: t2.micro

Status: Active

Instance State: Fulfilled

Persistence: One-time

The Spot Savings Summary also showed:

Total Spot Instances: 1

Savings: 67%

The screenshot displays the AWS Management Console interface for Spot Requests in the us-east-1 region. The left sidebar shows the navigation menu with 'Spot Requests' selected. The main content area is titled 'Spot Requests (1)' and contains a table with one request. Below the table is a 'Spot savings summary' section.

Request ID	Request type	Instance type	State	Capacity	Status	Persistence	Created
sir-2nszgh8n	instance	t2.micro	active	i-0c5439...	fulfilled	one-time	2 minute...

Spot savings summary			
A high-level summary of your savings across all of your running and recently terminated Spot Instances. For detailed reporting on your account-level Spot usage, visit Cost Explorer .			
Total Spot Instances	Total Spot cost (USD)	Total usage	Average cost per usage (USD)
1 instances	\$0.00 You saved 67%	1 vCPU hours 1 Memory(GiB) hours	\$0.0038 per vCPU hour \$0.0038 per Memory hour

7. Enable termination policy on the EC2 created in Task 2.

Task: Enable Termination Protection on the EC2 Created in Task 2

Instance: ubuntu-nginx-userdata

Instance ID: i-0b57adda2b93c4624

Steps

Opened AWS Console → EC2 → Instances.

Selected the Ubuntu instance created in Task 2.

Navigated to Actions → Instance settings → Change termination protection.

Selected Enable.

Clicked Save.

Verified the successful confirmation message.

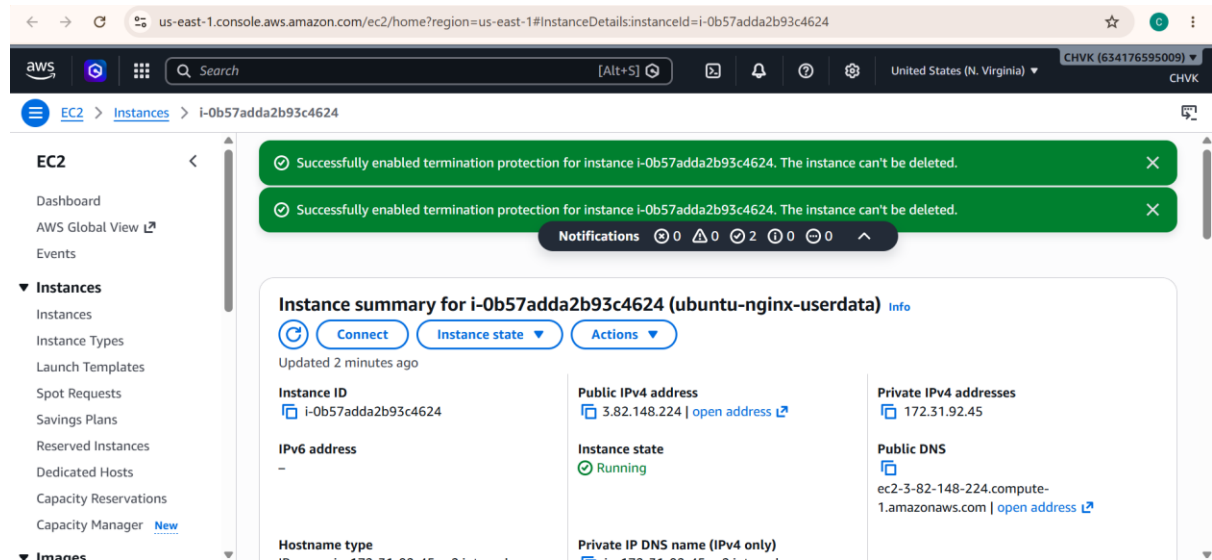
Verification

Successfully enabled termination protection for instance

i-0b57adda2b93c4624.

The instance can't be deleted.

Result Termination protection was successfully enabled for the Task 2 Ubuntu EC2 instance.



8. Launch one EC2 using AWS CLI.

Objective : To configure AWS CLI, launch an Amazon EC2 instance using AWS CLI, verify the instance status, and connect to the instance using SSH.

Step 1: Verify AWS CLI

```
aws --version
```

AWS CLI was installed and verified successfully.

Step 2: Configure AWS CLI

```
aws configure
```

The following details were configured:

AWS Access Key ID

AWS Secret Access Key

Default region: us-east-1

Default output format: json

Step 3: Verify AWS Credentials

```
aws sts get-caller-identity
```

The command successfully displayed the IAM user and AWS account information.

Step 4: Find the Amazon Linux AMI

```
aws ec2 describe-images --owners amazon --filters "Name=name,Values=amzn2-ami-hvm-2.0.*-x86_64-gp2" "Name=state,Values=available" --query 'Images | sort_by(@,&CreationDate)[-1].[ImageId,Name]' --output table
```

AMI used:

ami-0355fe804a20b1627

AMI Name:

amzn2-ami-hvm-2.0.20260817.0-x86_64-gp2

Step 5: Check Security Groups

```
aws ec2 describe-security-groups --query 'SecurityGroups[*].[GroupId,GroupName]' --output table
```

The required Security Group was selected with SSH access on port 22.

Step 6: Launch EC2 Instance

```
aws ec2 run-instances --image-id ami-0355fe804a20b1627 --instance-type t2.micro --key-name ec2-task-key --security-group-ids sg-085794a029eafca2d --count 1 --tag-specifications 'ResourceType=instance,Tags=[ {Key=Name,Value=cli-ec2-server}]'
```

The EC2 instance was successfully launched.

Step 7: Verify EC2 Instance

```
aws ec2 describe-instances --filters "Name=tag:Name,Values=cli-ec2-server" --query "Reservations[.Instances[.].[InstanceId,State.Name,PublicIpAddress,InstanceType]]" --output table
```

Result:

Instance ID: i-0a346cc78e29b177a

State: running

Public IP: 44.201.243.128

Instance Type: t2.micro

Step 8: Connect Using SSH

The .pem key file was stored in the Downloads folder.

Permission was changed:

```
chmod 400 ~/Downloads/ec2-task-key.pem
```

Then SSH connection was established:

```
ssh -i ~/Downloads/ec2-task-key.pem ec2-user@44.201.243.128
```

The connection was successful and the following prompt was displayed:

```
[ec2-user@ip-172-31-89-2 ~]$
```

Result

The EC2 instance was successfully launched using AWS CLI, verified as running, and accessed through SSH using the EC2 key pair.

```

chvka@karthik MINGW64 ~/OneDrive/Desktop
$ aws --version
aws-cli/2.36.30 Python/3.14.6 Windows/11 exe/AMD64

chvka@karthik MINGW64 ~/OneDrive/Desktop
$ aws configure

Tip: You can deliver temporary credentials to the AWS CLI using your AWS Console session by running the command 'aws login'.

AWS Access Key ID [None]: AKIAZHJ664RAVORN4BS3
AWS Secret Access Key [None]: VaseIdAjxyChm/RnjhRRNER8fqjyOoVOW5/UJow
Default region name [None]: us-east-1
Default output format [None]: json

chvka@karthik MINGW64 ~/OneDrive/Desktop
$ aws sts get-caller-identity

aws: [ERROR]: An error occurred (ParamValidation): argument operation: Found invalid choice 'get-caller-identity'

Maybe you meant:

  * get-caller-identity

usage: aws [options] <command> [<subcommand> ...] [parameters]
To see help text, you can run:

  aws help
  aws <command> help
  aws <command> <subcommand> help

chvka@karthik MINGW64 ~/OneDrive/Desktop
$ aws sts get-caller-identity
{
  "UserId": "ATDAZHJ664RAZK6NES5SQE",
  "Account": "634176595009",
  "Arn": "arn:aws:iam:634176595009:user/devops-cl1-user"
}

chvka@karthik MINGW64 ~/OneDrive/Desktop
$ aws ec2 describe-images --owners amazon --filters "Name=name,Values=amzn2-ami-hvm-2.0.*-x86_64-gp2" "Name=state,Values=available" --query 'Images | sort_by(&,&CreationDate)[-1].[ImageId,Name]' --output table
+-----+-----+
| DescribeImages |
+-----+-----+
| ami-0355fe804a20b1627 |
| amzn2-ami-hvm-2.0.20260817.0-x86_64-gp2 |
+-----+-----+

chvka@karthik MINGW64 ~/OneDrive/Desktop
$ aws ec2 describe-security-groups --query 'SecurityGroups[*].[GroupId,GroupName]' --output table
+-----+-----+
| DescribeSecurityGroups |
+-----+-----+
| sg-085794a029eafca2d | default |
| sg-0523f93313b713963 | launch-wizard-8 |
| sg-09b04031235da527c | launch-wizard-7 |
| sg-04b4eeca48a78266a9 | launch-wizard-5 |
| sg-09d73646f38c903db | launch-wizard-1 |
| sg-03d9f7229c8d67382 | launch-wizard-3 |
| sg-039a7cfa6876c0f1e | launch-wizard-4 |
| sg-093591809a7c74ef7 | launch-wizard-6 |
| sg-0ca9f348d863fa889 | launch-wizard-2 |
+-----+-----+

chvka@karthik MINGW64 ~/OneDrive/Desktop
$ aws ec2 describe-security-groups --group-ids sg-0857948029eafca2d --query 'SecurityGroups[0].[GroupId,GroupName,IpPermissions]' --output json
aws: [ERROR]: An error occurred (InvalidGroupId.Malformed) when calling the DescribeSecurityGroups operation: Invalid id: "sg-0857948029eafca2d"

chvka@karthik MINGW64 ~/OneDrive/Desktop
$ aws ec2 describe-security-groups --group-ids sg-0857948029eafca2d --query "SecurityGroups[0].[GroupId,GroupName,IpPermissions]" --output json
aws: [ERROR]: An error occurred (InvalidGroupId.Malformed) when calling the DescribeSecurityGroups operation: Invalid id: "sg-0857948029eafca2d"

chvka@karthik MINGW64 ~/OneDrive/Desktop
$ aws ec2 describe-security-groups --group-ids sg-085794a029eafca2d --query "SecurityGroups[0].[GroupId,GroupName,IpPermissions]" --output json
{
  "sg-085794a029eafca2d",
  "default",
  [
    {
      "IpProtocol": "tcp",
      "FromPort": 81,
      "ToPort": 81,
      "UserIdGroupPairs": [],
      "IpRanges": [
        {
          "CidrIp": "49.206.60.61/32"
        }
      ],
      "Ipv6Ranges": [],
      "PrefixListIds": []
    }
  ],
  {
    "IpProtocol": "tcp",

```

